

- 7 (a) (i) Specific latent heat of a substance is defined as the amount of heat released or absorbed when a unit mass of the substance undergoes a change of state at constant temperature. **B1**
- (ii) Energy is needed to do work against the atmosphere. **B1**
Difference in potential energies between molecules in the liquid state and gaseous state is larger than between solid and liquid state. **B1**
 (change of phase)
- (iii) 1. Heat gain = $m_i c_i [0 - (-7)] + m_i l_f + m_i c_i (T - 0)$
 Heat loss = $m_w c_w (85 - T)$

$$= (250 \times 10^{-3}) (4200) (85 - T)$$
 C1
 Heat gain = heat loss

$$(5 \times 5 \times 10^{-3}) (2100) (7) + (5 \times 5 \times 10^{-3}) (334 \times 10^3) + (5 \times 5 \times 10^{-3}) (4200) T = (250 \times 10^{-3}) (4200) (85 - T)$$
 C1

$$T = 70^\circ\text{C} \text{ (69.7 }^\circ\text{C)}$$
 A1
- (iii) 2. The specific heat capacity of the cup is not taken into consideration. **B2**
 Heat lost due to evaporation is negligible.

- (b) (i) the increase in internal energy of the system is equal to the sum of the heat supplied to the system and the work done on the system. **B2**

- (b) (ii) 1. Work done on = $p \times \Delta V$

$$= (0.75 \times 10^5) (5.0 - 1.6) \times 10^{-2}$$

$$= 2550 \text{ J} = 2.55 \text{ kJ}$$

M1

A1

- (b) (ii) 2.

Change in state	Process	Work done on gas / kJ	Heating on gas / kJ	Change in internal energy / kJ
A to B	Adiabatic	(1) -5.16	0	(2) -5.16
B to C	Isobaric	(3) 2.55	-4.83	(4) -2.28
C to D	Isothermal	0.43	(5) -0.43	(6) 0.00
D to A	Isochoric	(7) 0.00	7.44	(8) 7.44

A4

1 mark for correct values for (1) & (2)

1 mark for correct values for (3) & (4)

1 mark for correct values for (5) & (6)

1 mark for correct values for (7) & (8)

Using $\Delta U = Q + W_{on}$

(3) Value from (b)(ii)1.

$$(4) \Delta U_{B \text{ to } C} = -4.83 + (3) = -4.83 + 2.55 = -2.28 \text{ J}$$

$$(6) \Delta U_{C \text{ to } D} = 0 \text{ J}$$

Isothermal => constant temperature => no change in internal energy for ideal gas

$$(5) Q_{C \text{ to } D} = (6) - (0.43) = -0.43 \text{ J}$$

$$(7) W_{D \text{ to } A} = 0 \text{ J (Isochoric)}$$

$$(8) \Delta U_{D \text{ to } A} = (7) + 7.44 = 0 + 7.44 = 7.44 \text{ J}$$

(2) Since it is cyclic process, change in internal energy will be zero.

$$\Delta U = \Delta U_{A \text{ to } B} + \Delta U_{B \text{ to } C} + \Delta U_{C \text{ to } D} + \Delta U_{D \text{ to } A}$$

$$\Delta U_{D \text{ to } A} = -5.16 \text{ J}$$

$$(1) W = (2) - 0 = -5.16 \text{ J}$$

(ii) 3. Net work done on gas = $-5.16 + 2.55 + 0.43$
 $= -2.18 \text{ kJ}$

M1

A1

(ii) 4. Net work done by gas = $+2.18 \text{ J}$

$$\varepsilon = \frac{2.18}{7.44} \times 100\%$$
$$= 29.3\%$$

M1

A1